

```

silence_frames = win.getframerate()/2
# mono 16bit sound frames
silence_data = silence_frames * "\x00\x00"
wout.writeframes(silence_data)
data = win.readframes(win.getframerate())
wout.writeframes(data)
wout.writeframes(silence_data)
# divide the number of frames by frame rate
duration = float(wout.getframerate())/wout.getframerate()
win.close()
wout.close()
return duration

```

The code to get a wave file of the speech is a mere two lines. You can use the `espeak` or the `text2wave` command from the `Festival` package. The latter's voice quality is better. (I needed the frequency of the wave file to be 44100 for the sound for various scenes to be synchronised after conversion to MP3 audio.)

You can use the wave module to improve the presentation by inserting short silences at the start and the end of the wave file. This makes the presentation sound more natural.

End with a little animation

On the final image, a red square moves from left to right with 'The' written on it. You also create a green circle that moves from right to left with 'End' written on it. The two merge at the centre. The image is frozen for a second. You add the logot sound of the desktop to the scene. As it is the final image, you can ignore differences in the duration of the sound file and the video.

```

def animated_scene(bg_image):
    """A square with 'the' and a circle with 'end'
    float across a background image from opposite sides
    and merge.
    """
    duration = 2
    nframes = 25 * duration
    box_size = (100, 100)
    x_step = (640 - 100) / (2 * nframes)
    # create a red square 100x100
    im_square = Image.new('RGB', box_size)
    draw_s = ImageDrawDraw(im_square)

```

```

draw_s.rectangle([(0, 0), box_size], fill='RED')
draw_s.text((25, 25), 'The')
# create a green circle with diameter 100
im_circle = Image.new('RGB', box_size)
draw_c = ImageDrawDraw(im_circle)
draw_c.ellipse([(0, 0), box_size], fill='GREEN')
draw_c.text((25, 75), 'End')
# create a mask to show only the (green) circle
r_mask = im_circle.split()
# create the frames
scene_id = '99'
x_s = 0
x_c = 640 - 100
for frame_no in range(nframes - 1):
    image = bg_image.copy()
    image.paste(im_square, (x_s, 200))
    image.paste(im_circle, (x_c, 200), r_mask)
    x_s += x_step
    x_c -= x_step
    image.save(scene_id + "%03d" % frame_no + ".jpg")
# freeze the final frame for a second
for frame_no in range(nframes, nframes + 25):
    image = bg_image.copy()
    image.paste(im_square, (270, 200))
    image.paste(im_circle, (270, 200), r_mask)
    image.save(scene_id + "%03d" % frame_no + ".jpg")
# convert the frames into a scene. Use the logot sound
os.system('mencoder -audiofile /usr/share/sounds/logout.wav \
-oac mp3lame -mf /f + scene_id + *.jpg -mf fps=25 -o out.' \
+ scene_id + '.avi -ovc lavc -lavopts vcodec=mpeg4')

```

Even a small bit of animation takes some code. The core concept is that you create the foreground images that will appear to move. Make a copy of the original image that will serve as the background. Paste the foreground images at a new location and save the resulting image as a frame.

In our restless world, time is at a premium. So, go ahead and create 30-second just-in-time tutorials for your application, and they are sure to be a hit. 

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